The Schematics have been analyzed from
a die picture by InfosecDJ.
The chip is identified as:

MSM OKi
79H041

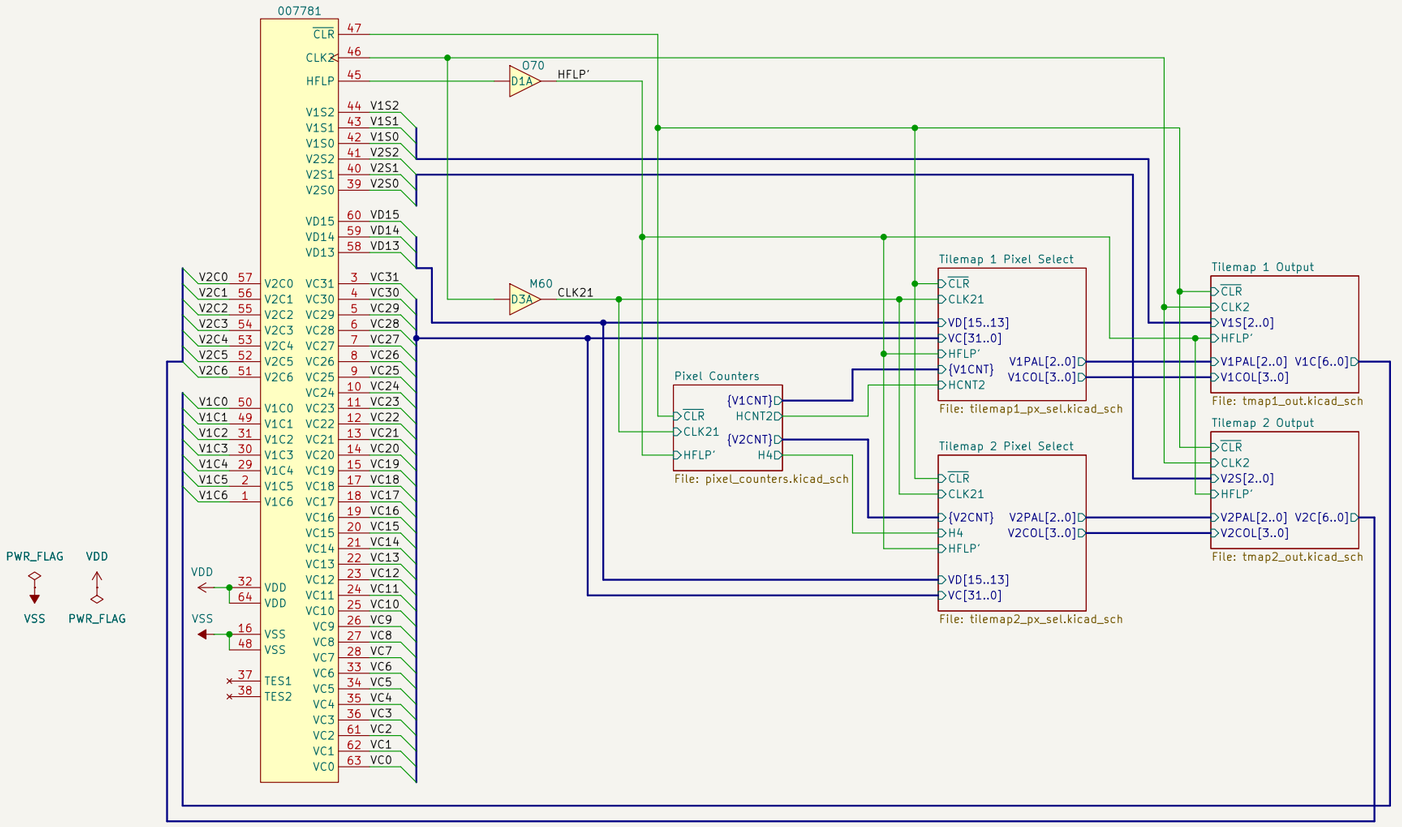
It is a MSM79V000 gate array
from the MSM70V000 series.
- 3289 Basic Cells
- 23 Columns, 143 Rows

The konami 007781 is a tilemap pixel shift register
with scroll. The tiles are 8x8 pixels, but the 007781
don't care about vertical line position. 007781 handles
two tilemaps at a time:
- V1, tilemap 1
- V2, tilemap 2

8 pixels, 4 bits each, of color data at a time are latched
into the registers. The palette code is latched 8 pixel
clocks earlier from VRAM. This is because the
data in the 007780 DRAM controller and DRAM takes 8
pixel clocks to process.

V1 latches pixels when the horizontal counter, going
from 0 to 7, goes to 2 (HCNT2) and V2 latches when
the horizontal counter goes to 4.
Data latched for:
- V1, when HCNT[2:0] == 2
- V2, when HCNT[2:0] == 4

The pixel code and palette color data are shifted
out between 1 to 9 pixel clocks after HCNT[2:0] == 4,
depending on if scroll or horizontal flip are active.

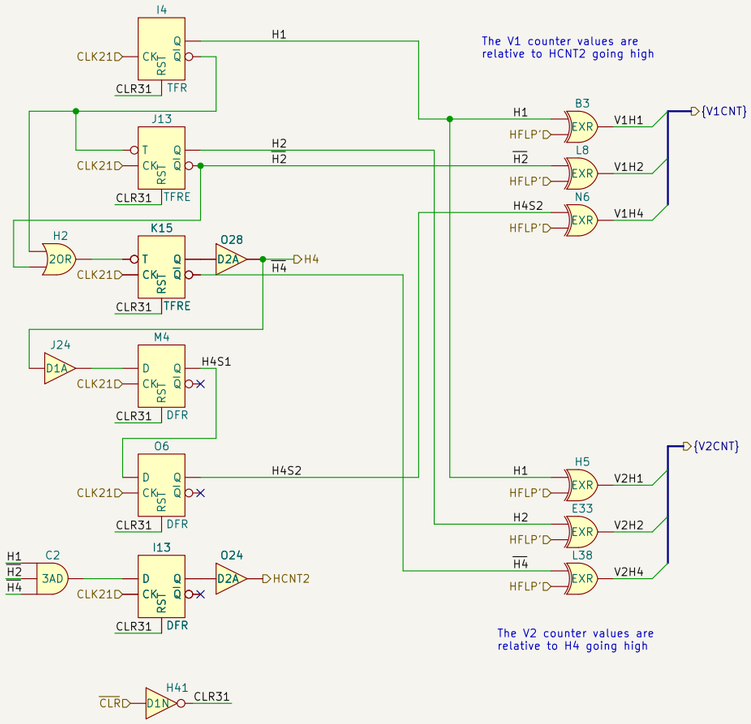


Ulf Skutnabba, twitter: @skutis77

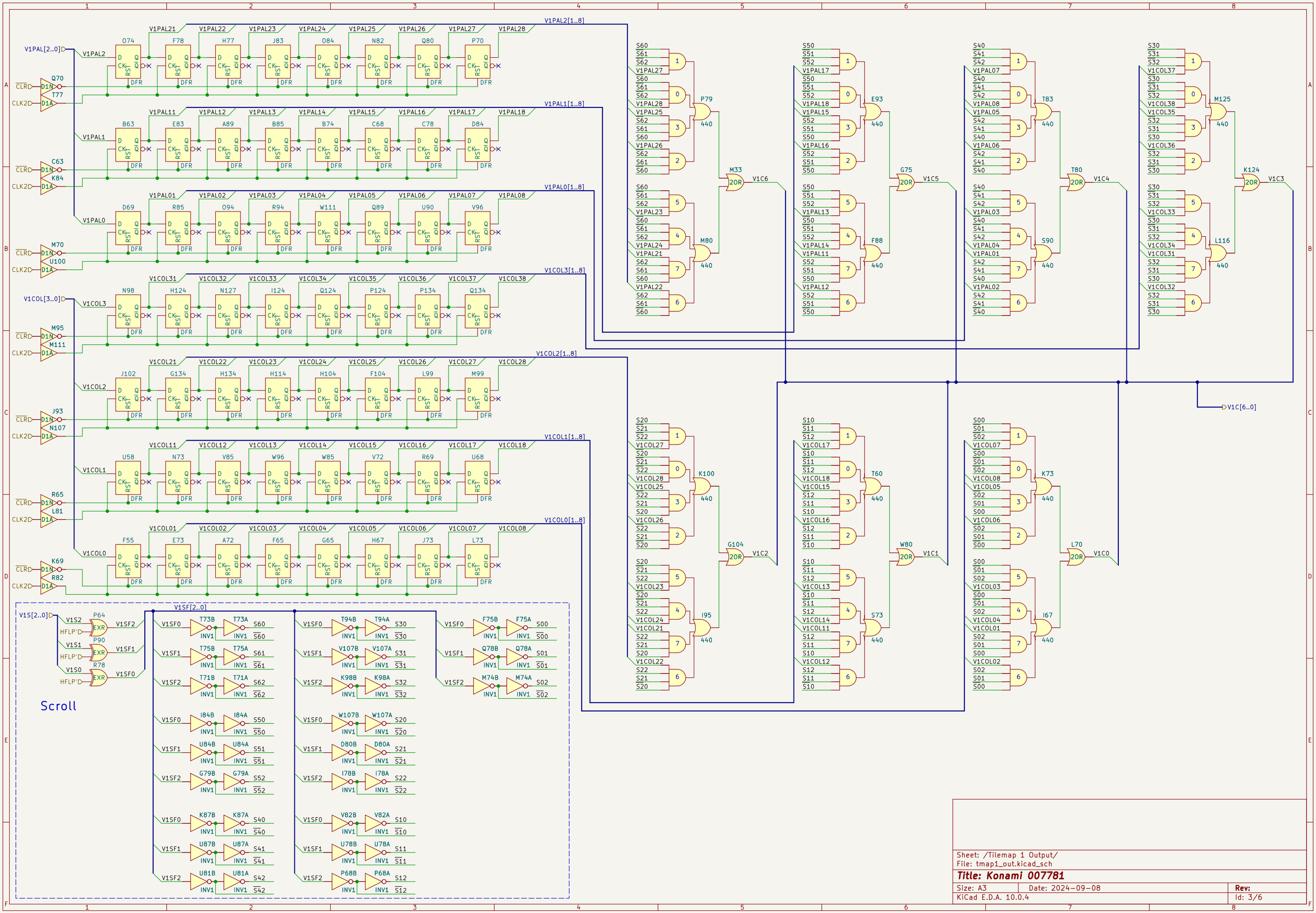
Sheet: /
File: 007781.kicad_sch

Title: Konami 007781

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/6



Sheet: /Pixel Counters/		
File: pixel_counters.kicad_sch		
Title: Konami 007781		
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/6



Sheet: /Tilemap 1 Output/
File: tmap1_out.kicad_sch

Title: Konami 007781

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 3/6

V2PAL[2..0]
V2PAL2
V2PAL1
V2PAL0
V2COL[3..0]
V2COL3
V2COL2
V2COL1
V2COL0
V2PAL2[1..8]
V2PAL1[1..8]
V2PAL0[1..8]
V2COL3[1..8]
V2COL2[1..8]
V2COL1[1..8]
V2COL0[1..8]
V2C[6..0]
V2SF[2..0]
V2S[2..0]
Scroll
K132
K128
J132
S112
T96
R110
F131
F126
E131
T122
T118
R122
N116
N110
O119
E118
D117
C112
G92
J98
H92
V2C6
V2C5
V2C4
V2C3
V2C2
V2C1
V2C0

The pixel data is latched when
HCNT2 goes high.
The pixel data order is

1. VC31,VC30,VC29,VC28
2. VC27,VC26,VC25,VC24
3. VC23,VC22,VC21,VC20
4. VC19,VC18,VC17,VC16
5. VC15,VC14,VC13,VC12
6. VC11,VC10,VC9,VC8
7. VC7,VC6,VC5,VC4
9. VC3,VC2,VC1,VC0

The palette color is delayed 8 pixels. It is because
the 007780 takes 8 pixels to process the pixel data.

Sheet: /Tilemap 1 Pixel Select/	
File: tilemap1_px_sel.kicad_sch	
Title: Konami 007781	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 10.0.4	Rev: Id: 5/6

The pixel data is latched when
HC4 goes high.
The pixel data order is

1. VC31,VC30,VC29,VC28
2. VC27,VC26,VC25,VC24
3. VC23,VC22,VC21,VC20
4. VC19,VC18,VC17,VC16
5. VC15,VC14,VC13,VC12
6. VC11,VC10,VC9,VC8
7. VC7,VC6,VC5,VC4
8. VC3,VC2,VC1,VC0

When flip is not active
pixel data is delayed one pixel.

Sheet: /Tilemap 2 Pixel Select/		
File: tilemap2_px_sel.kicad_sch		
Title: Konami 007781		
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/6